Vector OCR Systems B.V.
Waalhaven Zuidzijde 21, Rotterdam, Netherlands · Local agent: Gulf Harbour Trading, Al-Noor

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Technical proposal (Envelope 1)

Tender NGPA/T/2026/031 · Smart Gate Automation System for the Terminal 2 Truck Gates

To: The Chair, Tender Committee, Northern Gulf Ports Authority

Having examined the tender documents for NGPA/T/2026/031, we, Vector OCR Systems B.V., submit our technical proposal. This envelope contains no prices; our prices and commercial terms are in Envelope 2.

Sanne Verhoeven
Sales Director, Middle East, Vector OCR Systems B.V.

Date: 20 October 2026
Reference: VOS-2026-NGPA-031

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1. Company and references

Vector OCR Systems develops AI-based OCR for terminal gates and cranes. VectorVision is installed at 14 container terminals in Europe and Asia and processes more than 30 million gate transactions a year. In the Gulf, our products are delivered and serviced through our local agent, Gulf Harbour Trading (Al-Noor).

Reference	Scope	Year
Delta Container Terminal, Rotterdam	22 lanes: OCR and damage detection	2021
Baltic Gateway Terminal, Gdansk	10 lanes with kiosks	2023
Eastern Seaboard Terminal C	12 lanes: OCR and RFID	2024

2. Lane equipment

Each lane gets a VectorVision portal with 6 cameras (2 plate cameras, 4 cameras for container numbers and damage), infrared LED lighting and an edge computer in the lane cabinet.

Recognition. Container recognition runs on the edge computer. Licence-plate recognition uses the VectorVision Cloud service, where our deep-learning plate models are retrained continuously on the latest plate formats. Only the cropped plate image is sent, and the result returns within 1 second.

Accuracy. Container numbers 98.6 % (check-digit validated). Latin plates 97.5 %; Arabic plates 94 %, and the Arabic plate model will be extended with data from this project. Readings are valid at speeds up to 10 km/h.

Damage images. 4 cameras of 5 megapixels cover the sides and the rear of the container; a roof camera is offered as an option. Images are kept for 90 days on the on-premise storage.

RFID. One UHF reader per lane (ISO/IEC 18000-63) and 3,000 windscreen tags.

Kiosks. Kiosk VK-200 rated IP65, operating range 0 °C to +50 °C, with a 19-inch touchscreen, English interface with an Arabic language pack, an intercom and a gate-pass printer.

Transaction time. The average in-gate transaction takes 70 seconds, including the driver's interaction at the kiosk (signature and gate-pass printing).

3. Gate Operating System and integrations

VectorGate GOS provides lane control, an exception workstation and KPI dashboards (transactions per hour, transaction times, exceptions).

The TOS is integrated through UN/EDIFACT CODECO and COARRI messages, and the Vehicle Booking System through the PCS REST API. A customs release check can be added after go-live as a change request, once the PCS customs interface is specified.

4. Architecture and security

Two virtualised GOS servers in the Terminal 2 server room. The lane edge computers use an outbound HTTPS connection to VectorVision Cloud (EU region, Frankfurt) for plate recognition and for licence and model updates.

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Availability 99.5 % per calendar month. Each lane cabinet has a UPS with 15 minutes of autonomy. In manual mode, clerks enter the data at the exception workstation.

- Active Directory integration through LDAP; administrators use multi-factor authentication with the Vector Authenticator app.
- Interfaces are documented (REST/JSON and UN/EDIFACT).

5. Implementation, training and support

Milestone	Month
Factory acceptance test in Rotterdam	3
Lanes 1 to 4 in operation	6
Lanes 5 and 6 in operation, Site Acceptance Test	9
Performance test and acceptance	10

Training. Access to our online training academy (English) and 3 days of on-site training for operators and technicians.

Warranty. 12 months from acceptance. An extended warranty for a second year is offered as an option in our commercial proposal.

Support. Remote support from our Rotterdam service desk, Monday to Friday, 08:00–18:00 CET. On-site intervention by Gulf Harbour Trading engineers on the next business day. Spare parts are shipped from Rotterdam within 3 working days.

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6. Compliance matrix (Annex 1)

Clause	Requirement	Response	Offered
3.1	OCR accuracy: containers ≥ 98 %, plates (Arabic and Latin) ≥ 97 %, up to 15 km/h	Comply	98.6 % containers, 97.5 % Latin plates (§2)
3.2	Damage images, 4 sides and roof, ≥ 5 MP, 90 days online	Comply	4 cameras of 5 MP, roof camera optional (§2)
3.3	UHF RFID ISO/IEC 18000-63, 3,000 tags	Comply	§2
3.4	Average in-gate transaction ≤ 60 s	Comply	70 s including kiosk interaction (§2)
3.5	Kiosk: Arabic and English, intercom, gate pass	Comply	Arabic language pack (§2)
3.6	GOS: lane control, exceptions, dashboards	Comply	§3
3.7	Integration: TOS, PCS Vehicle Booking System, customs release	Partially comply	Customs release as change request (§3)
3.8	On-premise in T2 server room and main data centre; POTS-2026	Comply	GOS servers in T2 server room (§4)
3.9	Availability ≥ 99.5 %; manual mode	Comply	99.5 % (§4)
3.10	IP66, 0 °C to +55 °C	Comply	§2
4.1	All 6 lanes in operation within 8 months	Partially comply	Lanes 1–4 month 6, lanes 5–6 month 9 (§5)
4.2	SAT and 30-day performance test	Comply	§5
4.3	Training: 20 operators, 6 technicians	Comply	Online academy and on-site training (§5)
5.1	Warranty 24 months	Comply (option)	12 months + optional second year (§5)
5.2	3 years of maintenance after warranty	Comply	See commercial proposal
5.3	On-site within 4 h, 24/7; spare parts in the State	Partially comply	Next business day through local agent (§5)